A Quantum Gauge Theory of Time (QGT): Foundations from a Noetherian Balance

Pasquale Camelia[1*]

[1*]Independent Researcher, Italy.

Corresponding author(s). E-mail(s): pasquale.camelia@gmail.com;

Abstract

This work derives time without postulating it. Starting from a single conserved Noetherian flow under restricted accessibility, we show that global conservation under restricted accessibility requires an irreducible compact internal cycle, that observables must arise as reduced bilinear invariants, and that representability forces a non-compact four-dimensional domain. The product $\mathbb{R}^{\mathbf{4}} \times \boldsymbol{S}^{\mathbf{1}}$ is not assumed, but emerges as the unique representational structure capable of simultaneously supporting conservation, observability, and compensation under loss. The factor $\boldsymbol{S}^{\mathbf{1}}$ is not an additional spatial dimension: it is the minimal topological support for the globally conserved coherent phase, which remains inaccessible to projected observables. Irreversible projection then forbids cyclic equivalence and forces a unique partial ordering of structural recurrences. Time is identified with this ordering relation itself, not with a parameter or dimension. What persists as matter is what survives loss; what orders loss is time.

Keywords: Foundations of physics, Problem of time, Noether theorem, Gauge symmetry, Temporal ordering, Irreversibility, Spinorial structure

1 Introduction

The status of time in fundamental physics remains conceptually unsettled. While quantum mechanics and quantum field theory treat time as an external parameter governing evolution, general relativity incorporates temporal and spatial components within a dynamical geometric structure. Despite their empirical success, these frameworks do not share a common *operational definition* of time, nor do they provide a unified account of temporal ordering, scale separation, and the origin of material observables *without introducing time itself as a primitive element.*

This tension is not merely interpretational. It appears concretely in the absence of a preferred temporal variable in generally covariant systems, in the coexistence of reversible microscopic laws with irreversible macroscopic behaviour, and in the lack of a structural mechanism linking temporal ordering to the organisation of physical observables. In relativistic quantum field theory, time enters as a coordinate of spacetime rather than as a dynamical carrier endowed with its own conserved structure; in gravitational theories, temporal and spatial components are treated on equal geometric footing, yet no intrinsic account of temporal discreteness, ordering, or accessibility of internal phase is provided.

Across these domains, symmetries and conservation principles play a central role. Gauge theories derive their consistency and predictive power from internal symmetries, while Noether's theorem establishes the relation between continuous symmetries and conserved currents [1]. In standard formulations, however, Noether conservation constrains the dynamics of fields evolving *with respect to* time; it does not address whether temporal structure itself can arise from a conservation requirement.

The approach taken here starts from a single fact: a conserved flow associated with a continuous internal symmetry. Without a conserved quantity identifiable across symmetry-related configurations, no notion of persistence exists, independently of any assumed temporal parameter. The sole condition imposed is

$$\partial_I J^I = 0\,, \tag{1}$$

where I labels a domain left intentionally unspecified — not presupposed to be spacetime, not carrying a metric or equation of motion. The question is what structures become necessary once access to this flow is incomplete. Time is not postulated. Each section introduces structure only where consistency requires it.

2 The conserved flow as sole input

The sole physical input is the continuity condition

$$\partial_I J^I = 0,$$

where J^I are the components of a conserved flow and I labels a domain left deliberately unspecified — not assumed to be spacetime, not assumed to carry a metric. This is not a law governing dynamics. It is the weakest condition under which the identity of a conserved quantity can be consistently maintained under redistribution. Conservation is taken as the starting point; admissible structures, and only then admissible dynamics, follow from it.

The condition becomes non-trivial when access to the flow is restricted: when only a subset of its invariants is operationally available. What structures must then emerge to keep conservation meaningful? No discreteness, dimensionality, or carrier structure is assumed. Each enters only when required.

3 Phase compactness

The continuity condition introduced in Section 2 constrains admissible physical descriptions without specifying dynamics or geometry. Its nontrivial content emerges once the conserved flow is considered globally, rather than pointwise. In particular, the existence of a transportable conserved quantity implies the presence of a **global mode** associated with the continuous internal symmetry.

This global component — the **zero mode** of the conserved flow — is the sector of the current invariant under local redistributions while encoding global transport. It is the minimal carrier for the continuity condition to be satisfied across symmetry-related configurations.

A defining property of any continuous internal symmetry admitting a globally transportable generator is the existence of an associated phase parameter. The connected one-dimensional Lie groups are, up to isomorphism, $\mathbb{R}$ and S^1. The group $\mathbb{R}$ carries no intrinsic return map: repeated transport along $\mathbb{R}$ never identifies any two configurations, so no finite holonomy class exists and the zero mode fails to define a globally conserved carrier. The group S^1 is the unique compact connected one-dimensional Lie group; it admits closed orbits, finite holonomy, and a well-defined return map under repeated transport. Consistency therefore selects

$$\chi \in S^1 \tag{2}$$

as the minimal phase structure for a globally transportable zero mode. No further one-dimensional option exists.

No temporal interpretation is attached to χ. It labels symmetry-related configurations of the conserved flow and secures the coherence of their global identification — nothing more. Under restricted accessibility, this compact phase becomes the point at which incompleteness acquires structural significance.

4 Domain separation

The structures introduced so far—the Noetherian continuity condition and the existence of a compact phase associated with a globally conserved zero mode—are sufficient to ensure the consistency of conservation at the global level. They are not sufficient to guarantee that the same conservation condition remains well defined once access to the conserved flow becomes restricted.

When the conserved flow is globally accessible, the Noetherian continuity condition

$$\partial_I J^I = 0\,.$$

can be enforced without ambiguity. Under restricted accessibility, however, not all components of the conserved flow can be operationally retained. In particular, components associated with the internal phase of the zero mode cease to be accessible, while phase-independent invariants remain available.

For the continuity condition to remain well posed, the admissible domain must decompose into two structurally distinct sectors,

$$D = D_{\mathrm{coh}} \oplus D_{\mathrm{red}}. \tag{3}$$

Here D_{coh} denotes the sector in which the full zero-mode structure, including the compact phase, is accessible, while D_{red} denotes the complementary sector in which only reduced, phase-independent invariants of the conserved flow can be retained.

Without this separation, conservation would be violated or operationally meaningless in the reduced sector.

The notion of locality here is not spatial. It refers to the weakest relational structure required to define redistribution within each sector — adjacency between components of the conserved flow, independently of any metric.

Within the coherent sector D_{coh}, the zero mode can be transported with full phase information preserved, and conservation is enforced at the level of the complete internal symmetry. Within the reduced sector D_{red}, phase information is inaccessible, and conservation constrains only the redistribution of phase-independent quantities. The same continuity condition therefore admits distinct structural implementations across the two sectors.

The separation of domains introduces an intrinsic asymmetry. This asymmetry is not dynamical and does not rely on geometry. It arises from the requirement that a single conservation principle remain valid under restricted accessibility. No interpretation is attached to the two sectors in terms of spacetime regions, dimensions, or physical substrates.

Global conservation, compact phase structure, and incomplete access now jointly constrain the carrier itself.

5 The spinorial carrier

The domain decomposition $D = D_{\mathrm{coh}} \oplus D_{\mathrm{red}}$ constrains the admissible carriers of the conserved flow. The carrier must support a compact internal phase, admit a non-trivial projection when phase information is lost, and preserve the single continuity condition across both sectors. Together these exclude all but one class of representations.

A scalar carrier is excluded. A scalar transforms as $\phi \mapsto e^{i\alpha}\phi$ under $U(1)$; once the phase $e^{i\alpha}$ is projected away, only $|\phi|^2$ survives — a single real number carrying no internal relational structure. The coherent and reduced sectors become identical, making domain separation vacuous.

A vector carrier is likewise excluded. Vectors transform under compact symmetries in a single-valued way: transport by 2π around S^1 returns the vector to the same configuration, $V(\chi + 2\pi) = +V(\chi)$. Under projection the phase can be absorbed into a gauge choice, leaving reduced invariants that are again trivial — no sign ambiguity survives to distinguish the two sectors. Single-valuedness is precisely what must fail for the coherent/reduced distinction to be non-trivial.

Among compact representations, the minimal representation that remains non-trivial under loss of phase information is therefore one that is **double-valued under**

2π **transport**. This property is topological rather than dynamical: it reflects the structure of the representation space under the action of the compact symmetry.

Let Ψ denote the minimal carrier of the conserved flow. The defining property of this representation is

$$\Psi(\chi + 2\pi) = -\,\Psi(\chi). \tag{4}$$

This relation expresses the double-valuedness of the carrier under closed transport of the compact phase $\chi \in S^1$. Single-valued representations would collapse to trivial invariants under projection, eliminating any structural distinction between coherent and reduced sectors and rendering the domain separation introduced in Section 4 physically vacuous.

Here the term *spinorial* is used in a strictly structural sense. It refers to the topology of the representation space under the action of the compact internal symmetry, and does not refer to spin in spacetime, Lorentz representations, or particle degrees of freedom. No assumption regarding dimensionality, metric structure, or relativistic kinematics is introduced at this stage.

The spinorial character of the carrier is essential for restricted accessibility to have physical content. In the coherent sector D_{coh}, the full phase structure of Ψ is accessible, and the sign ambiguity under closed transport is resolved by the internal phase. In the reduced sector D_{red}, where phase information is inaccessible, the carrier can only be accessed through phase-independent bilinear combinations, which remain invariant under $\chi \to \chi + 2\pi$:

$$\mathcal{O} = \Psi^\dagger M \Psi. \tag{5}$$

Here M denotes a generic element of the internal algebra acting on the carrier. The operator M plays no dynamical or spectral role here: it labels admissible bilinear forms that survive projection. Only the existence of such bilinears is required; their detailed classification will be addressed later. These bilinears constitute the minimal observables compatible with projection and continuity. Their role is purely structural: they represent the weakest form of accessibility compatible with global conservation once phase information is lost.

The necessity of a spinorial carrier therefore follows directly from the requirement that a single conservation principle remain valid under compact phase structure and restricted accessibility. Without double-valuedness under closed transport, no non-trivial reduced sector could exist, and the domain separation required by conservation would be structurally empty.

The identification of a minimal spinorial carrier completes the hierarchy of constraints initiated by the Noetherian continuity condition: global conservation enforces compact phase structure; restricted accessibility enforces domain separation; and the coexistence of both enforces a spinorial representation as the minimal admissible carrier. In the next section, the internal symmetry structure associated with this carrier follows from the same consistency requirements.

Proposition 5.1

Let the conserved coherent flow be represented by a single-valued carrier on the compact internal cycle. Then any observable surviving projection is necessarily phase-blind and collapses into a single equivalence class.

As a consequence, no non-invertible projection can be consistently defined, and the distinction between coherent and reduced domains becomes operationally trivial.

Therefore, the internal carrier cannot be single-valued. It must be double-valued, transforming as

$$\Psi(\chi + 2\pi) = -\,\Psi(\chi)\,,$$

in order for global conservation and restricted accessibility to coexist non-trivially. The spinorial structure is thus not assumed, but **forced by consistency of the projection**.

6 Internal symmetry

The identification of a minimal spinorial carrier in Section 5 constrains the admissible internal symmetry under which the conserved flow is transported. The symmetry must be compatible with the entire hierarchy established so far: compact phase structure, domain separation under restricted accessibility, and preservation of a single Noetherian continuity condition across coherent and reduced sectors.

Three requirements are therefore imposed on the internal symmetry:

1. it must admit a **compact phase** associated with the globally conserved zero mode;
2. it must act **non-trivially** on a spinorial carrier under closed transport;
3. it must remain **structurally stable under restricted accessibility**, allowing non-trivial reduced invariants after projection.

Purely Abelian internal symmetries are insufficient. Groups such as $U(1)$, or equivalently real Abelian groups such as $SO(2)$, admit a single generator associated with compact phase rotations. Their action on a spinorial carrier is single-channel and does not support an internal relational structure. When access to the phase is lost, the reduced sector collapses to trivial invariants, eliminating any structurally meaningful distinction between coherent and reduced domains.

To preserve a non-trivial reduced organisation, the internal symmetry must therefore admit **multiple interrelated generators** acting on the spinorial carrier, such that loss of access to the compact phase does not trivialise the conserved flow. These generators must transform into one another under the symmetry action itself; otherwise, projection would isolate independent conserved channels, contradicting the requirement of a single conserved flow.

The minimal compact Lie group satisfying these requirements is SU(2). Among compact non-Abelian Lie groups, the smallest is SU(2): it has rank 1, dimension 3, and its fundamental representation is the minimal pseudo-real spinorial representation of complex dimension two — precisely the double-valued carrier forced by Section 5. Any smaller compact non-Abelian group does not exist: SU(2) is the floor. Any larger group — $SO(3)$ admits only integer-spin representations and lacks a faithful spinorial one; SU(3) and above introduce additional invariant channels, contradicting the single-flow condition. SU(2) is therefore not chosen but singled out by the joint constraints.

At the level of the coherent domain, the internal structure is not factorised. The compact phase associated with the zero mode and the non-Abelian SU(2) structure act jointly on a continuous set of symmetry-related configurations. Prior to projection, there is no meaningful separation between an SU(2) sector and a $U(1)$ sector: the two are structurally entangled within the conserved carrier.

Under restricted accessibility, this continuous organisation cannot be maintained. Preservation of the Noetherian flow requires the identification of equivalence classes of configurations rather than individual continuous orbits. In this way, the internal SU(2) structure reorganises into an effectively non-continuous internal organisation compatible with incomplete access.

The residual internal organisation accessible after projection is therefore characterised by the coset structure

$$\mathrm{SU}(2)/U(1). \tag{6}$$

This follows because restricted accessibility does not remove internal transformations, but identifies configurations related by transformations that become operationally indistinguishable. The appropriate mathematical structure encoding such identification is therefore a quotient by the stabiliser subgroup.

Here $U(1)$ appears as the stabiliser of the compact phase associated with the zero mode. The coset notation does not represent a physical space or configuration manifold, nor a geometric reduction, but an internal symmetry structure encoding the organisation of the conserved flow once continuity is no longer accessible.

The restriction to a single globally conserved Noetherian flow further excludes internal product structures, which would generically define independent invariant channels and therefore multiple conserved currents.

Among compact non-Abelian groups compatible with a real spinorial carrier, restricted accessibility, and a single conserved flow, SU(2) is therefore the minimal admissible internal structure. Larger groups or direct products introduce additional invariants not supported by the imposed constraints and are excluded by minimality.

In the next section, this structure will be shown to require an extension of the algebraic framework supporting the spinorial carrier, leading to a spectral constraint.

7 Clifford extension

7.1 Why Cl(1, 3) is insufficient

Sections 5–6 establish that the conserved Noetherian flow is carried by a **real spinorial structure** endowed with a **non-Abelian internal symmetry**. This carrier must support non-trivial internal organisation under restricted accessibility while preserving a *single* continuity condition.

Let $\{\gamma_\mu\}$ with $\mu = 0, \ldots, 3$ generate the Clifford algebra $\mathrm{Cl}(1,3)$, defined by

$$\{\gamma_\mu, \gamma_\nu\} = 2\,\eta_{\mu\nu}\mathbf{1} \tag{7}$$

Here $\mathbf{1}$ denotes the identity operator on the spinorial carrier, while $\eta_{\mu\nu}$ denotes the bilinear form fixing the signature of the Clifford algebra. No spacetime, metric geometry, or Lorentzian structure is assumed or implied at this stage; the choice of $\mathrm{Cl}(1,3)$ is purely algebraic and serves only to fix the internal anticommutation relations of the spinorial carrier.

Within $\mathrm{Cl}(1,3)$ there exists, up to normalisation, a unique element that anticommutes with all γ_μ,

$$\gamma_5 \equiv i\,\gamma_0\gamma_1\gamma_2\gamma_3, \qquad \{\gamma_5,\gamma_\mu\} = 0, \tag{8}$$

with fixed square

$$\gamma_5^2 = \pm\mathbf{1}. \tag{9}$$

As a consequence, $\mathrm{Cl}(1,3)$ admits **only one independent internal grading** compatible with spinorial structure. Any further element anticommuting with all γ_μ is proportional to γ_5 and therefore introduces no additional algebraic direction.

However, Section 6 showed that preservation of a non-trivial reduced organisation under projection requires a **real non-Abelian internal structure**. The unique chirality element γ_5 is insufficient: it merely partitions the carrier space into two conjugate sectors and does not encode an internal relational organisation among generators.

Therefore, the algebraic structure supporting the conserved flow **cannot be exhausted by** $\mathrm{Cl}(1,3)$. An extension by at least one additional Clifford generator is required.

A single additional generator γ_6 is introduced, defined by

$$\{\gamma_6,\gamma_\mu\} = 0, \qquad \mu = 0,1,2,3, \tag{10}$$

and algebraically independent of γ_5.

No assumption is made on the spectrum or normalisation of γ_6. Its role is purely structural: it provides the **minimal internal algebraic direction** required to represent the non-Abelian organisation of the conserved carrier. No spacetime interpretation is attached to γ_6.

The problem is now to determine **how this extended structure must close** once continuous phase accessibility is lost.

The internal symmetry identified in Section 6 is SU(2), whose generators satisfy

$$[T_i, T_j] = \varepsilon_{ijk}\,T_k. \tag{11}$$

This non-commutativity implies that internal transport depends on ordering. Here the term *torsion* refers exclusively to the non-commutativity of internal transport in SU(2), not to spacetime torsion or affine geometry.

Under full access to the compact phase, this ordering dependence can be continuously compensated: different transport paths remain connected by smooth reparametrisations.

Under restricted accessibility, however, the phase information required to compensate this non-commutativity is no longer available. A **residual internal effect** remains. This residual cannot vanish: if it did, the internal symmetry would collapse to an Abelian one, contradicting Section 6.

Therefore, preservation of a *single* Noetherian continuity condition requires that the residual internal organisation **close under repeated internal transport**, even when continuous phase compensation is lost.

7.2 Spinorial closure

The spinorial carrier established in Section 5 satisfies

$$\Psi(\chi + 2\pi) = -\,\Psi(\chi). \tag{12}$$

This sign is not incidental. It means that traversing the compact cycle once returns the carrier with a global sign reversal. If λ is an eigenvalue of the transport operator for one traversal, then the inverse traversal — the same cycle in the opposite orientation — acquires the sign from the spinorial condition and contributes $-\lambda^{-1}$, not $+\lambda^{-1}$. The spinorial double-valuedness therefore forces the conjugate pair to be $(\lambda, -\lambda^{-1})$.

Under restricted accessibility, the phase information distinguishing the two orientations is lost. The orientation-invariant combination of the pair $(\lambda, -\lambda^{-1})$ is their sum:

$$K = \lambda + (-\lambda^{-1}) = \lambda - \lambda^{-1}. \tag{13}$$

By Vieta's formulae, the minimal polynomial with $(\lambda, -\lambda^{-1})$ as its roots — sum $= K$, product $= -1$ — is:

$$\lambda^2 - K\lambda - 1 = 0. \tag{14}$$

The product of the roots being -1 is not an assumption: it is forced by the spinorial sign $\Psi(\chi + 2\pi) = -\Psi(\chi)$, which makes the inverse traversal contribute $-\lambda^{-1}$ rather than $+\lambda^{-1}$.

The degree 2 is the minimal degree encoding the full conjugate pair. A degree-1 polynomial retains only one orientation. A polynomial of degree $2p > 2$ factors into p independent conjugate pairs, each defining a separate conserved channel and contradicting the single-flow condition.

The compact internal cycle $S^1_\chi(\tau)$ is rank-one by construction, carrying a single globally conserved mode λ. The self-consistency condition — that the fixed point λ^* of the ratio v_{k+1}/v_k satisfies the same closure equation — requires $\lambda^* = 1 + (\lambda^*)^{-1}$, i.e. (14) at $K = 1$. Since no dimensional parameter has been introduced, K must be a positive integer, and $K = 1$ is the unique value for which the fixed point is real, non-degenerate, and free of external parameters:

$$v_{k+2} = v_{k+1} + v_k, \tag{15}$$

the unique closure compatible with a rank-one spinorial carrier, single-flow conservation, and the spinorial sign constraint.

Remark 1 (Consistency with the dilation operator of Section 9) The Pincherle operator $D = x\,d/dx$ has eigenfunctions x^λ. Under restricted accessibility, which identifies x^λ and $x^{-\lambda}$ as two readings of the same representative modulo the spinorial sign, the accessible spectral combination is $\lambda - \lambda^{-1}$, consistent with K as defined above. The closure equation (14) with $K = 1$ is therefore the projection-invariant spectral equation shared by the spinorial transport of this section and the dilation structure of Section 9 — two readings of the same underlying constraint.

7.3 The spectral invariant and operator

Equation (14) with $K = 1$ gives the minimal polynomial

$$\lambda^2 - \lambda - 1 = 0. \tag{16}$$

Its two real roots are

$$\lambda_\pm = \frac{1 \pm \sqrt{5}}{2}. \tag{17}$$

These roots encode directional information associated with the internal transport prior to projection. Under restricted accessibility, however, such directional information is not representable in the reduced sector. Projection eliminates sensitivity to the orientation of transport along the internal cycle and retains only quantities invariant under inversion.

For a second-order closure, the only inversion-invariant information carried by the pair of roots is their symmetric spectral separation. Explicitly, while the individual eigenvalues $\lambda_\pm$ are not accessible after projection, their difference remains invariant:

$$\lambda_+ - \lambda_- = \sqrt{5}. \tag{18}$$

Accordingly, preservation of the Noetherian continuity condition under projected non-Abelian transport forces the appearance of an internal operator whose spectrum consists of two symmetric real values,

$$\mathrm{spec} = \{\pm\sqrt{5}\}. \tag{19}$$

The numerical value $\sqrt{5}$ is therefore not introduced as a fundamental constant. It arises uniquely as the discriminant of the minimal second-order closure polynomial compatible with non-Abelian residual transport, conservation of a single global flow, and closure under restricted accessibility. First-order closures collapse to trivial or periodic structures, while higher-order closures introduce additional independent parameters and violate minimality.

The internal spectral invariant must act linearly on the same carrier space that supports the conserved flow. The minimal way to encode such an operator within the extended algebra is to identify it with the new Clifford generator γ_6,

$$\gamma_6 \Psi = \pm\sqrt{5}\,\Psi, \qquad \gamma_6^2 = 5\,\mathbf{1}. \tag{20}$$

Any alternative realisation would require additional generators or free parameters, contradicting the minimality established in Sections 1–6.

This identification completes the algebraic structure:

- γ_6 is required because $\mathrm{Cl}(1,3)$ is insufficient;
- its spectrum is fixed by closure under projected non-Abelian transport;
- no additional assumptions are introduced.

The next section uses this structure to formulate the Noether balance between coherent and reduced sectors, and to derive the discrete mode organisation that survives projection.

8 Balance and discrete modes

For any admissible codimension-one cut Σ of the domain, one may define the associated invariant

$$Q[\Sigma] \equiv \int_{\Sigma} J^I \, d\Sigma_I. \tag{21}$$

By virtue of $\partial_I J^I = 0$, the quantity $Q[\Sigma]$ is invariant under continuous deformations of Σ. Q is the only conserved quantity the theory requires.

8.1 Failure of local continuity

Restricted accessibility is implemented by a projection map Π acting on the conserved flow. The accessible representative is

$$j^I \equiv \Pi[J^I]. \tag{22}$$

The projection Π does not commute with non-Abelian internal transport. The full flow satisfies continuity; its projected representative does not:

$$\partial_I J^I = 0, \qquad \partial_I j^I \neq 0. \tag{23}$$

This inequality does not signal a violation of conservation, but the loss of access to part of the conserved flow. The non–commutativity between projection and internal SU(2) transport is the origin of the imbalance between what is accessible and what remains coherent but inaccessible.

Because the total invariant Q remains conserved, the failure of local continuity for j^I must be compensated by a complementary contribution that is not representable in the projected sector. Denoting by

$$Q_{\rm acc}[\Sigma] \equiv \int_{\Sigma} j^I \, d\Sigma_I \tag{24}$$

the accessible part of the invariant, the complementary contribution $Q_{\rm coh}$ is defined implicitly by the requirement that the total charge remain invariant,

$$Q = Q_{\rm acc} + Q_{\rm coh}. \tag{25}$$

Because the non–commutativity described above is structural, changes in $Q_{\rm acc}$ between two admissible cuts cannot vanish in general. Conservation of Q therefore implies the **Noether balance law**

$$\Delta Q_{\rm acc} + \Delta Q_{\rm coh} = 0. \tag{26}$$

This balance is not a definition but a consequence of the incompatibility between projection and non–Abelian transport: any loss in the accessible sector is exactly compensated by the coherent complement.

8.2 Discreteness

Section 7 showed that, once continuous phase accessibility is lost, non–Abelian internal transport cannot be compensated infinitesimally. There exists no infinitesimal generator compatible with both restricted accessibility and preservation of a single conserved flow.

As a result, the residual organisation of the projected sector cannot be described by a continuous evolution parameter. The only admissible representation of change is through **successive closure steps** of the internal organisation.

Let $k \in \mathbb{Z}$ label these successive closures. The Noether balance must therefore be realised stepwise:

$$\Delta Q_{\text{acc}}^{(k)} = -\Delta Q_{\text{coh}}^{(k)}. \tag{27}$$

The index k does not represent time; it labels successive minimal closure steps required to maintain Noetherian consistency after projection.

8.3 Holonomy classes

Independently of the iteration index k, the compact internal phase implies the existence of closed internal transport. Let $\mathcal{H}$ denote the internal holonomy associated with one full loop of the compact phase. For a spinorial carrier,

$$\mathcal{H}^n = -1, \qquad \mathcal{H}^{2n} = 1, \tag{28}$$

for some integer $n \geq 1$.

The integer n labels **inequivalent global holonomy classes**. It is distinct from the iteration index k:

- k labels successive closure steps of the projected organisation;
- n labels inequivalent global classes of internal transport.

The special case

$$n = 0 \tag{29}$$

denotes the trivial projected sector, in which no non–trivial holonomy information survives projection.

8.4 Admissible sectors

The framework admits a **single globally conserved flow**. If a holonomy class labelled by n were reducible, $n = ab$ with $a, b > 1$, the corresponding transport would factor into independent sub–cycles. Under projection, such a factorisation would generically give rise to independent conserved channels, in contradiction with the single–flow condition.

Irreducibility under projection therefore excludes generic composite factorisations as stable representatives of the reduced organisation. The **simplest admissible non–trivial sectors** are consequently associated with **primitive values of** n, together with the low–order sectors enforced by spinorial symmetry,

$$n \in \{0, 1, 2, 3, 5, 7, 11\}. \tag{30}$$

Higher values are not excluded in principle, but are not required for the minimal organisation analysed here. The present classification concerns only the lowest irreducible sectors relevant for stable projected organisation.

The explicit set above is presented as a minimal low–sector catalogue used in this paper; it is not asserted as the complete admissible spectrum.

This classification does not constitute a number–theoretic claim. It identifies **minimal admissible classes** compatible with irreducibility, restricted accessibility, and preservation of a single conserved Noetherian flow. No statement is made regarding the exhaustive set of admissible values, nor regarding the exclusion of all composite sectors in extended frameworks.

The projection of a single Noetherian flow has therefore produced, without introducing time or metric structure, a single conserved invariant Q, a balance law between accessible and coherent sectors, a necessarily discrete representation of change indexed by k, and a classification into irreducible holonomy classes labelled by n, with a distinguished trivial sector $n = 0$. What the discrete organisation cannot yet do is support quantitative comparison between its inequivalent representatives. That requirement is the subject of the next section.

9 Scale

Projection induces a discrete organisation of inequivalent representatives, generated by successive closure steps and classified by global holonomy classes. A further question remains open:

> What minimal structure is required to compare inequivalent projected representatives while preserving the identity of a single conserved flow?

The answer is a scale structure. No spacetime, metric, dynamics, or phenomenological notion of scale is assumed.

The dilation operator employed here traces back to Pincherle's operational calculus [2], see also the modern historical and mathematical analysis in [3]. It is introduced here purely as a structural operator of comparability, not as a dynamical or physical generator.

Comparability arises from the mismatch between the inaccessible continuous closure of the conserved flow and the discrete family of inequivalent closures that survive projection.

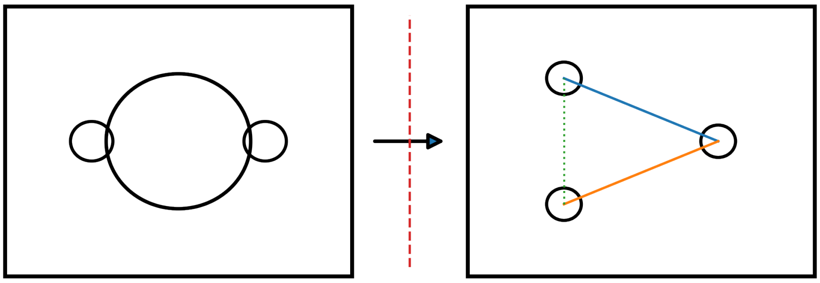

Fig. 1 Structural decomposition induced by restricted accessibility. The small circle on the left represents the globally conserved coherent sector of the Noetherian flow, including the compact internal phase that becomes inaccessible under projection. The small circle on the right represents the projected accessible sector, consisting exclusively of reduced bilinear invariants. The projection map (Π) is non-invertible and does not preserve local closure. The multiple discrete elements shown within the accessible sector represent distinct reduced invariant structures that remain simultaneously observable after projection. Their plurality indicates that the accessible domain is non-trivial but incomplete, and that compensation for lost coherence occurs through a finite set of inequivalent invariants, rather than through a continuous parameter or a single observable. The large enclosing circle represents the global Noetherian balance, in which the conserved invariant is maintained only as the sum of coherent and accessible contributions, even though neither sector is independently conserved or complete. The diagram encodes structural relations only and does not represent geometry, dynamics, or temporal ordering.

9.1 From ordering to scale

Let Π_k denote the sequence of projected representatives obtained through successive closure steps k, as constructed in Section 8. These representatives are ordered in the sense that they are not mutually equivalent under the internal symmetry action.

However, ordering alone does not define a scale. An ordering provides only relational information, such as precedence or inclusion, but does not allow quantitative comparison between inequivalent representatives.

In particular, partial orders, graphs, or purely discrete labels distinguish representatives without assigning relative magnitude.

Ordering alone does not define scale. If no scale structure exists, the only available relation between inequivalent representatives Π_k and $\Pi_{k'}$ is qualitative — one can establish precedence but not ratio. A map preserving the conserved flow modulo a multiplicative factor then has nowhere to act: it either collapses to an equivalence (identifying what was assumed inequivalent) or is undefined. A scale structure is therefore not optional; it is what allows the conserved flow to survive comparison between inequivalent representatives.

Finite hierarchies fail: they introduce privileged endpoints, incompatible with unbounded iterative closure. Discrete labels encode distinction but not magnitude. Partial orders and monotone maps provide relational structure but no ratios. None satisfies simultaneously single-flow conservation, unbounded iteration, and quantitative comparability.

9.2 Dilation and the Pincherle generator

The only remaining possibility is a structure that relates inequivalent representatives by relative magnitude,

$$\Pi_k \longrightarrow \lambda\, \Pi_k, \qquad \lambda > 0.$$

Such transformations form a dilation structure. This dilation does not represent physical stretching, geometric size, or renormalisation flow. It is the minimal algebraic structure required to compare representatives while preserving the identity of the conserved flow.

To represent dilations in a minimal, real, parameter-free manner, we seek a single generator D satisfying:

- linearity,
- absence of any intrinsic scale,
- compatibility with unbounded iteration,
- preservation of the identity of the conserved flow.

The natural candidate is the standard generator of scale transformations, well known in analysis and representation theory, defined by

$$D = x\,\frac{d}{dx}.$$

This is the unique first-order linear operator implementing a dilation without introducing additional parameters or dimensional constants.

In the terminology of operational calculus it coincides with what is often referred to as the Pincherle operator [2]. Its eigenfunctions x^{λ} come in conjugate pairs $(x^{\lambda}, x^{-\lambda})$ under restricted accessibility, since projection identifies the two transport orientations established in Section 7. The accessible spectral combination is therefore $\lambda - \lambda^{-1}$, the same exchange invariant (13) that fixes the spinorial closure of Section 7 to the Fibonacci recursion (15). The Pincherle operator is thus not chosen independently: it is the unique generator whose spectral structure under projection is consistent with the closure already forced by the spinorial carrier. In the present framework, however, it plays no dynamical role: it is used solely as a structural generator of comparability between inequivalent projected representatives.

Any alternative realisation would either require additional generators or introduce explicit scale parameters, thereby violating the minimality constraints imposed by the theory.

9.3 Discrete levels, continuous generator

Although the hierarchy of representatives is discrete, indexed by k, their comparison requires a continuous generator. This does not introduce physical continuity or dynamics.

The continuous generator encodes relative magnitude between discrete levels, providing a faithful representation of a discrete hierarchy without introducing new degrees of freedom.

Scale arises as the minimal structure that allows inequivalent projected representatives of a single conserved flow to be quantitatively related.

10 Observables and matter

What survives as observable once coherence is no longer directly accessible? Observables, dimensional organisation, and matter emerge from conservation under restricted accessibility — not from additional postulates.

10.1 Reduced invariants

Under projection, the fundamental spinorial carrier introduced earlier is no longer directly accessible. What remains observable must therefore be constructed from quantities invariant under the loss of internal phase.

The minimal such quantities are bilinear invariants of the form

$$\mathcal{O} = \Psi^T C\, M\, \Psi. \tag{31}$$

Here Ψ denotes the underlying spinorial carrier constrained by the requirements derived in Section 5: neutrality of the conserved flow, absence of observable phase, and compatibility with real, phase-invariant quantities under projection.

The matrix C denotes the invariant bilinear form required to construct real scalar quantities from the spinorial carrier. Its role is purely structural: it encodes the involutive conjugation necessary for the definition of projection-stable bilinears and does not introduce additional degrees of freedom.

The operator M denotes a generic element of the internal algebra acting on the spinorial carrier. It labels distinct bilinear invariants but carries no dynamical or spectral interpretation. Only those choices of M that preserve reality and phase-independence under projection give rise to admissible reduced observables.

The above requirements uniquely enforce a real, self-conjugate spinorial structure. In the standard terminology, such a real spinor is referred to as a *Majorana spinor* [4]. The Majorana character is therefore not assumed as a physical hypothesis, but follows necessarily from the structural constraints imposed by restricted accessibility.

Such bilinears are (i) invariant under internal transport prior to projection, (ii) insensitive to the inaccessible phase, and (iii) stable under restricted accessibility.

No linear object survives projection without collapsing to triviality. Observables therefore do not represent fundamental degrees of freedom, but *reduced invariants* of a deeper coherent structure.

The set of admissible bilinear invariants constitutes a complete family of reduced observables, in the sense that any accessible quantity can be expressed as a linear combination of such invariants. This defines a structural resolution of the spinorial carrier, without introducing any additional topological, Hilbert, or spectral structure.

A "local" observable here is not defined by position but by how much of the coherent organisation survives projection — how restricted the accessibility is. Locality is structural, not spatial.

10.2 The $n = 5$ sector

The discrete organisation derived in Section 8 admits holonomy sectors labelled by the integer n, each corresponding to a distinct balance pattern between accessible and coherent components of the conserved flow.

Stability of reduced observables requires more than the mere existence of bilinear invariants. It requires that their **norm remain real, bounded, and non-degenerate under discrete balance iteration** induced by restricted accessibility.

Let $\mathcal{O}_n$ denote the reduced bilinear invariant associated with sector n. Structural stability demands that:

1. $|\mathcal{O}_n|$ be real and non-vanishing;
2. $|\mathcal{O}_n|$ be invariant under inversion of internal orientation;
3. $|\mathcal{O}_n|$ be preserved under successive balance steps without collapse or degeneracy.

The second-order closure derived in Section 7 fixes the residual spectral separation of the projected internal operator to

$$\mathrm{spec} = \{\pm\sqrt{5}\}.$$

The associated invariant bilinear norm is therefore fixed to

$$|\mathcal{O}|^2 = (\sqrt{5})^2 = 5.$$

The integer n labels the minimal closure length required for this invariant norm to be realised consistently within the discrete balance hierarchy. The identification of the equilibrium sector follows from a fixed-point condition on the Fibonacci closure (15): the boundary closes — that is, Fibonacci growth F_n coincides with the available linear capacity n — precisely when

$$F_n = n. \tag{32}$$

Direct inspection of the Fibonacci sequence gives $F_1 = 1$, $F_2 = 1$, $F_3 = 2$, $F_4 = 3$, $F_5 = 5$; for $n \geq 6$, Fibonacci growth is exponential in φ while linear capacity grows by 1, so $F_n > n$ strictly. The unique non-trivial solution is $n = 5$.

For $n < 5$: the Fibonacci growth has not yet reached the capacity threshold; the invariant norm $|\mathcal{O}|^2 = 5$ cannot be realised without collapse of the bilinear structure under iteration. For $n > 5$: growth exceeds capacity; additional closure steps assign non-zero weight to operationally inaccessible directions, introducing redundancy without new independent invariants.

The sector $n = 5$ is therefore the unique **structural equilibrium**: the configuration in which Fibonacci closure capacity exactly matches the spectral norm imposed by non-Abelian transport, realised once and without degeneracy.

In this precise sense, $n = 5$ is the minimal spinorial zero mode of the projected organisation: the lowest sector in which the bilinear norm matches the spectral constraint of Section 7.

Projection is not a neutral operation. Each step of restricted accessibility removes a portion of the coherent organisation of the conserved flow.

Information arises here as a **compensatory organisation**: it is the only means by which the system can preserve identifiable structure once coherence is no longer directly accessible.

This compensation is necessarily finite. As the discrete hierarchy progresses:

- informational organisation initially stabilises observables,
- but inevitably becomes insufficient,
- leading to structural collapse at $n = 1$.

The sector $n = 1$ therefore represents the failure of compensation: coherence can no longer be defended by information. The subsequent sector $n = 0$ does not represent annihilation. The conserved flow itself is not destroyed; rather, it re-enters the cycle without memory, stripped of all internal organisation.

The result is a cycle:

$$\text{coherence} \;\rightarrow\; \text{compensation} \;\rightarrow\; \text{collapse} \;\rightarrow\; \text{reset}.$$

No notion of temporal evolution has been assumed. Yet a static description has become impossible.

10.3 The representability threshold

The bilinear observables identified above must admit a domain in which they can be stably represented without degeneracy. This requirement does not refer to embedding in spacetime, but to **representability**: the minimal structure capable of supporting non-degenerate reduced invariants.

Below a certain threshold, bilinear observables collapse before stabilisation; above it, representational freedom introduces redundancy without increasing stability. The minimal domain compatible with stable bilinear organisation and real Clifford structure is effectively **four-dimensional**.

This representational threshold is consistent with known structural properties of real Clifford algebras: non-degenerate real bilinear invariants associated with Majorana-type carriers first admit a minimal complete realisation in four effective dimensions.

Below this threshold, compensatory informational organisation collapses before stable observables can be sustained; above it, redundancy increases without structural gain. Four effective dimensions therefore arise as a **threshold of representability**, not as an assumed spacetime.

Matter, in this framework, is not a substance or excitation. It is what remains identifiable once coherence is no longer directly accessible — the residue of a conserved flow that persists despite irreversible erosion of its coherent structure.

No claim is made regarding spacetime geometry, gravitational dynamics, or quantum field interactions. "Matter" here denotes a structural class: the stable reduced invariants of a single conserved flow that remain identifiable under irreversible loss of coherence. The product $\mathbb{R}^4 \times S^1$ is not spacetime; geometric notions such as metric or causal order may be introduced at a later effective level, but are not present here.

The structure cannot be described statically. Coherence is preserved at the price of information; at $n = 1$ that price exceeds capacity and compensation fails; at $n = 0$ memory is erased while the conserved flow survives. The cycle restarts without record, yet not without structure. A static description is impossible — ordering is forced.

11 The representational structure $\mathbb{R}^4 \times S^1$

The preceding sections did not assume a background arena in which physics unfolds. Instead, they imposed a sequence of structural constraints:

- a single globally conserved Noetherian flow,
- restricted local accessibility of its coherent carrier,
- non-Abelian internal organisation,
- discrete closure under projection,
- compensatory informational organisation,
- comparability inducing scale without dynamics.

The problem is no longer interpretative. It is a problem of representational existence:

> Which manifolds can simultaneously realise all these constraints, and which are structurally impossible?

The question is answered by exclusion, not by construction.

11.1 The irreducible internal cycle

Sections 3–7 established that the conserved carrier possesses an internal phase with the following properties:

- it is globally transportable,
- it remains compact under closure,
- it is locally inaccessible under projection,
- it acts non-trivially on the coherent structure while remaining invisible to observables.

An unbounded parameter is excluded, since it would destroy global identifiability under repeated transport. Absorption into observable bilinears is excluded, since all local observables are phase-blind.

What remains is an irreducible internal cycle, required to preserve global coherence without observational access. The minimal topological support for such a cycle is one-dimensional and compact. No additional structure is permitted, and no weaker structure suffices.

The result is an internal factor S^1, which is neither temporal nor geometric, but the minimal carrier of conserved coherence under restricted accessibility.

11.2 Four dimensions

All locally accessible quantities arise, by construction, as bilinear invariants of the projected spinorial structure. This imposes a strict representational constraint: the extended domain must admit bilinears that are simultaneously

- real and non-degenerate,
- stable under Noether balance,
- compatible with restricted accessibility,
- capable of supporting compensatory informational organisation.

The real Clifford algebra $\mathrm{Cl}(p, q)$ with $p+q = d$ first admits a faithful real spinorial representation — a Majorana spinor — in dimension $d = 4$ with signature $(1, 3)$ or $(3, 1)$. Below four dimensions, no real matrix representation of the full algebra exists that simultaneously supports the double-valued carrier of Section 5, the non-Abelian structure of Section 6, and real non-degenerate bilinear invariants: the algebra is too small, and degeneracy or collapse follows. Above four dimensions, the algebra grows redundantly — new generators appear that carry no additional independent observable structure under the single-flow constraint, only duplicating existing invariants. Four dimensions are therefore the unique threshold where the Majorana carrier finds its minimal faithful real home.

Moreover, the extended domain cannot be compact. Compactness would enforce global identifications among projected observables, contradicting restricted accessibility and re-entangling coherence with the reduced sector.

We are thus forced to a non-compact four-dimensional domain, denoted $\mathbb{R}^4$, understood purely as a representational space, devoid of metric, causal, or dynamical assumptions.

The appearance of four effective dimensions as a representability threshold is consistent with the deep relation between spinorial structures and four-dimensional geometry emphasised in spinor-based formulations of relativistic physics [5].

11.3 Factorisation and recurrence

The internal cycle S^1 and the extended domain $\mathbb{R}^4$:

- were derived independently,
- encode fundamentally different structural roles,
- are subject to incompatible accessibility conditions.

Projection enforces their separation:

- the internal phase remains globally conserved but locally inaccessible,
- the extended domain supports only projected, bilinear observables.

Any non-trivial fibration would correlate internal phase accessibility with projected observables, violating this enforced separation. Twisted bundles and mixed structures are therefore excluded.

The representational structure is thus forced to factorise as

$$\mathbb{R}^4 \times S^1. \tag{33}$$

This factorisation is not imposed by symmetry breaking, compactification, or dimensional reduction. It is the minimal structure that exists at all under the constraints of the theory.

Remark 2 (Relation to Kaluza–Klein and higher-dimensional theories) The factor S^1 is not an additional spatial dimension in the sense of Kaluza–Klein compactification [6, 7]. In those theories, a compact extra dimension carries dynamical fields and generates gauge interactions through Fourier decomposition; its radius is a free parameter subject to experimental bounds. Here, by contrast, S^1 carries the **coherent phase of the conserved Noether current**, which is globally defined but **locally inaccessible to projected observables by construction**. It enters the structure not as a geometric dimension of spacetime, but as the minimal topological support required for global conservation under restricted accessibility. No fields propagate on S^1, no compactification radius is introduced, and no tower of Kaluza–Klein modes is generated. The product $\mathbb{R}^4 \times S^1$ is therefore a representational structure — the unique one compatible with the constraints of the theory — not an arena in which dynamics unfolds.

The resulting structure supports the following pattern:

- global conservation of coherence,
- local loss of phase accessibility,
- emergence of compensatory informational organisation,
- finite capacity of compensation,
- inevitable collapse,
- preservation of the conserved flow itself.

This organisation exhibits structural recurrence, not temporal evolution. The term "cycle" denotes recurrence of structural roles, not succession in time.

The theory therefore reaches a decisive point: a conserved flow persists through irreversible loss of coherence, but cannot be ordered without an additional principle.

11.4 Scope and limits

$\mathbb{R}^4$ is not spacetime. S^1 is not time. No metric, causal order, or dynamics has been introduced. What has been derived is the unique representational structure in which coherence can be conserved, observables can exist, information can compensate loss, and collapse does not annihilate the flow. Time has not been assumed. It has been made necessary. It will emerge as the minimal ordering compatible with the survival of a conserved flow under irreversible loss of coherence.

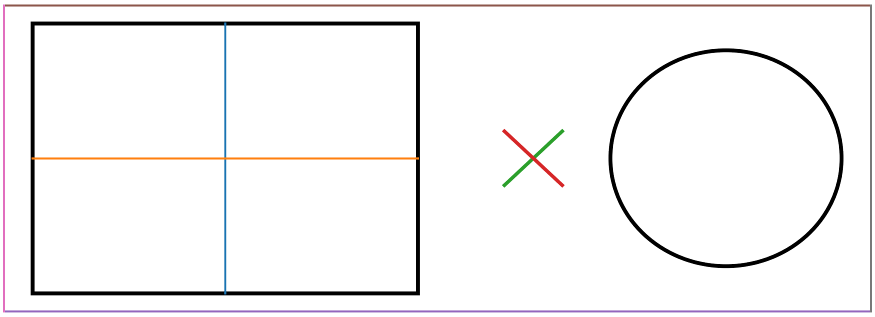

Fig. 2 Schematic representation of the coexistence of accessible and inaccessible structures after projection. The figure does not represent a geometry, coordinate system, or embedding. The crossed lines schematically indicate independent directions of locally accessible bilinear observables, insensitive to the internal phase and carrying no memory of global coherence. The circular loop represents the compact internal phase (S^1) required for global conservation of the coherent carrier, which remains inaccessible to projected observables. The relative placement of elements is purely illustrative and carries no spatial meaning. In particular, the separation between the loop and the crossed structure expresses incompatibility of accessibility, not geometric distance. The crossed symbol indicates the structural impossibility of reconstructing the internal phase from projected observables. Distinct smaller loops denote inequivalent structural recurrences of the conserved flow under projection, which cannot be identified or inverted despite global conservation.

12 Temporal ordering

The construction of Sections 3–11 yields structural recurrence: the conserved flow persists through coherence, compensation, collapse, and reset, without restoring memory. Nothing in this construction has introduced a criterion for ordering distinct recurrences. The question is whether one can be derived from what already exists, without adding a new primitive.

Recurrence alone does not yield ordering. A purely cyclic structure identifies successive instances modulo a loop action, treating them as equivalent — but the projection has already forced non-equivalence: the loss of coherent degrees of freedom cannot be undone. Cyclic identification would erase this irreversibility, collapsing inequivalent recurrences into a single class. It is therefore excluded.

Let R_i denote a structural recurrence: a complete instance of the balance between accessible and coherent sectors under the representational split of Section 11. The projected representative satisfies $\partial_I j^I \neq 0$; the deficit is carried by the coherent complement through the balance law; the reset at $n = 0$ preserves the flow but does not restore inaccessible structure. Successive recurrences are therefore generically inequivalent: no invertible internal transformation preserves accessibility between them. This asymmetry requires no metric, no dynamics, no external parameter — it follows from the projection alone.

The weakest ordering structure compatible with irreversible non-equivalence is a partial order. Define $R_i \prec R_j$ when R_j is reachable from R_i under projection-induced loss, with $a(R) = \dim Q_{\mathrm{coh}}(R)$ strictly decreasing: $a(R_j) < a(R_i)$. Antisymmetry follows immediately — $R_i \prec R_j$ and $R_j \prec R_i$ would require $a(R_j) < a(R_i)$ and $a(R_i) < a(R_j)$ simultaneously. Transitivity is the monotonicity of a. Non-totality reflects that recurrences with the same a but different internal organisation are structurally incomparable.

Any ordering that ignores loss collapses inequivalent recurrences. Any total ordering imposes fictitious global comparability excluded by restricted accessibility. Any ordering requiring auxiliary labels introduces a new primitive. The partial order induced by irreversible loss is the only admissible one.

Time is the name for that order. It is not a dimension, not a parameter, not a coordinate. It is the relation $\prec$ between structurally inequivalent recurrences of the same conserved flow — the unique ordering that restricted accessibility forces and cannot undo.

The structure $\mathbb{R}^4 \times S^1$ derived in Section 11 separates two incompatible roles: S^1 carries the compact coherent phase required for global transport, while $\mathbb{R}^4$ supports the persistent bilinear observables. Temporal ordering belongs to neither factor. It is not located in a coordinate of S^1 or $\mathbb{R}^4$ — it is a relation between structurally inequivalent recurrences of the same representational split, arising only when the two factors are linked by projection-induced loss.

Conclusions

The construction presented here starts from one structural fact — a globally conserved Noetherian flow under restricted accessibility — and asks what must follow without assuming spacetime, metric, or dynamics. The answer is a chain of forced consequences: a compact internal phase S^1, a spinorial carrier, non-Abelian internal organisation, a discrete balance between accessible and coherent sectors, a dilation structure for comparability, real bilinear observables requiring four extended dimensions, and a factorised representational structure $\mathbb{R}^4 \times S^1$. None of these is postulated. Each is the only option left after excluding what fails.

At the end of this chain, ordering becomes unavoidable. The conserved flow persists through irreversible loss of coherence, but successive recurrences cannot be related by any invertible transformation that preserves accessibility. A partial order is the weakest structure that encodes this non-equivalence without introducing a hidden parameter. Time is the name for that order.

This is not a claim about dynamics, interactions, or phenomenology. It is a claim about what must exist before any of those can be formulated. Dynamical realisations remain open; they are not needed to establish the necessity of the ordering.

If one accepts global conservation and restricted accessibility, temporal ordering cannot be avoided. It is not added to the structure. It is what remains when nothing else can be added.

Declarations

Funding

The author received no financial support for the research, authorship, and/or publication of this article.

Conflict of interest / Competing interests

The author declares that there are no competing interests.

Ethics approval and consent to participate

Not applicable. This study does not involve human participants, animals, or personal data.

Consent for publication

Not applicable.

Data availability

No datasets were generated or analysed during the current study.

Materials availability

Not applicable.

Code availability

No code was generated or used in the development of this theoretical work.

Author contribution

The author solely conceived, developed, and wrote the manuscript.

Appendix A — Symbolic realisation

A.1 The carrier field

To verify the internal consistency of the framework, we introduce a symbolic carrier field

$$\Psi(x, \chi)$$

defined on the representational structure

$$\mathbb{R}^4 \times S^1.$$

The variable $x \in \mathbb{R}^4$ labels the extended representational domain identified in Sections 10–11, while $\chi \in S^1$ parametrises the compact internal cycle required for global transport of coherence.

The carrier Ψ is taken to be **real and double-valued** under transport along the internal cycle,

$$\Psi(\chi + 2\pi) = -\,\Psi(\chi),$$

as derived in Section 5. This property is not associated with physical spin or particle degrees of freedom. It expresses the minimal algebraic structure required to maintain non-trivial domain separation under restricted accessibility.

No equation of motion is postulated for Ψ. The field serves only as a symbolic representative of the conserved coherent carrier discussed throughout the paper.

A.2 The conserved current

The foundational assumption of the theory is the existence of a **single globally conserved flow**, expressed abstractly by a continuity condition.

In representational terms, this can be written as a conserved current

$$J^I = \Psi^T\, \Gamma^I\, \Psi$$

satisfying

$$\partial_I J^I = 0, \qquad I = 1, \ldots, 5.$$

Here the matrices Γ^I provide a representation of the internal algebraic structure required to support the conserved flow. No Clifford metric or spacetime interpretation is assumed. The conservation law is **not derived from an action principle** and does not encode dynamics.

It is essential to stress that this current is not an observable current in the reduced domain. It represents the **primitive conserved quantity** whose persistence under restricted accessibility drives the entire construction.

No assumption is made that the set $\{\Gamma^I\}$ realises a metric Clifford algebra; the notation is purely representational and does not encode spacetime signature or causal structure.

A.3 Reduced observables

Restricted accessibility is implemented by a projection operator Π that eliminates direct dependence on the internal phase variable χ.

At the level of the carrier field, Π acts by suppressing phase-sensitive information, rendering linear expressions in Ψ operationally meaningless in the reduced sector. What survives projection are **real bilinear invariants** of the form

$$\mathcal{O} = \Psi^T\, C\, \mathcal{M}\, \Psi,$$

where C denotes the charge-conjugation structure and $\mathcal{M}$ acts on the internal algebraic indices.

Such bilinears are:

- invariant under internal transport prior to projection,

- insensitive to the inaccessible phase,
- stable under the balance relations derived in Sections 8–10.

The projection Π is **non-invertible** by construction. No operation within the reduced domain allows reconstruction of the suppressed coherent degrees of freedom. This formalises the notion of irreversible loss of accessibility used throughout the paper.

Although no Hilbert space structure is assumed here, the organisation of reduced bilinear invariants admits an interpretation as a structural resolution of the spinorial carrier, analogous to decomposition procedures familiar from functional analysis. This analogy is not required for the present construction and plays no role in the derivation.

The symbolic carrier $\Psi(x, \chi)$ introduced here is not an additional postulate. It is a notational convenience for verifying that the abstract structures of the main text admit a consistent algebraic realisation.

References

[1] Noether, E.: Invariante variationsprobleme. Nachrichten von der Königlichen Gesellschaft der Wissenschaften zu Göttingen, 235–257 (1918)

[2] Pincherle, S., Amaldi, U.: Le Operazioni Distributive e Le Loro Applicazioni all'Analisi. Zanichelli, Bologna (1901)

[3] Mainardi, F., Pagnini, G.: The role of salvatore pincherle in the development of fractional calculus. In: Coen, S. (ed.) Mathematicians in Bologna 1861–1960. Springer, Basel (2012)

[4] Majorana, E.: Teoria simmetrica dell'elettrone e del positrone. Il Nuovo Cimento **14**, 171–184 (1937)

[5] Penrose, R., Rindler, W.: Spinors and Space-Time, Volume 1: Spinor and Twistor Methods in Space-Time Geometry. Cambridge University Press, Cambridge (1986). Reprint 2008

[6] Kaluza, T.: Zum Unitätsproblem der Physik. Sitzungsberichte der Preussischen Akademie der Wissenschaften, 966–972 (1921)

[7] Klein, O.: Quantentheorie und fünfdimensionale relativitätstheorie. Zeitschrift für Physik **37**, 895–906 (1926)